A Diagnostic Methodology for Failure-Mode Ablation of SFM, TTC, and ORCA Pedestrian Steering Paradigms

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Keywords: Diagnostic Methodology, Minimum-Intervention Controls, Steering Ablation, Pedestrian Dynamics.

Abstract: We argue that failure-mode ablations of pedestrian steering paradigms should be accompanied by minimum-intervention controls that isolate each paradigm's operative mechanism from trivial alternatives, and we instantiate this methodology on the canonical Social Force (SFM), time-to-collision (TTC), and Optimal Reciprocal Collision Avoidance (ORCA) triad. A zone-decomposed ablation across bottleneck, bidirectional, and crossing geometries ($n = 25$ per cell) isolates two paradigm-specific failure modes: TTC anticipation does not break symmetric arching deadlocks at narrow exits (0/25 successful evacuations for SFM and SFM+TTC at $w = 0.8$ m, vs. 13–14/25 for ORCA-enabled configurations), and ORCA reciprocal yielding gridlocks counterflow. A minimum-intervention control–$C_1+\varepsilon$, the SFM-only baseline C_1 with a per-step Gaussian velocity perturbation ε–locates the deadlock-resolution threshold at $\sigma_{50} = 0.049$ m/s (95% CI $[0.044, 0.053]$); within $n = 25$, passive noise is indistinguishable from ORCA's linear-program (LP)-based velocity selection as a deadlock-breaking mechanism, demonstrating that a headline paradigm advantage can be attributed to passive symmetry-breaking rather than to distinctive machinery. Calibration against 4 776 Forschungszentrum Jülich (FZJ) trajectories and a held-out out-of-distribution (OOD) evaluation are reported. We position the audit principle as the methodological contribution; the SFM/TTC/ORCA diagnostic as its first application.

1 INTRODUCTION

Crowd disasters remain a persistent threat to public safety. The 2022 Itaewon crush killed 159 people in a narrow alley in Seoul during a Halloween gathering (Lee et al., 2024); the 2010 Love Parade disaster killed 21 at a tunnel bottleneck in Duisburg (Helbing and Mukerji, 2012). Both events share a common aetiology: localised pedestrian density exceeded survivable thresholds long before organisers detected the hazard, and passive crowd dynamics alone were insufficient to dissipate the buildup. Predictive tools that can anticipate dangerous density accumulations and evaluate mitigations *before* an event are therefore a prerequisite for evidence-based crowd safety management, and the reliability of such tools rests, ultimately, on the pedestrian steering model at their core.

Pedestrian steering paradigms are routinely credited with resolving specific failure modes of crowd simulation – lane formation in counterflow, arch dissolution at narrow exits, standoff resolution at intersections – and these credit assignments flow directly into design guidance, benchmark rankings, and production-system choices. We argue that such mechanism claims should be audited before they are relied upon. Concretely: whenever a paradigm X is credited with resolving a failure mode Y through a distinctive mechanism M, the claim should be tested against a minimum-intervention control that isolates M from trivial alternatives such as passive symmetry-breaking, thermal fluctuation, initial-condition heterogeneity, or numerical-artefact resolution. The methodological contribution of this paper is to articulate this audit principle and demonstrate, in a concrete instance, that it changes what we can conclude from an ablation.

We instantiate the methodology on the canonical Social Force (SFM) (Helbing and Molnár, 1995; Helbing et al., 2000), time-to-collision (TTC) (Karamouzas et al., 2014), and Optimal Reciprocal Collision Avoidance (ORCA) (Fiorini and Shiller, 1998; van den Berg et al., 2011) triad with a *zone-decomposed* failure-mode ablation – contact metrics reported per spatial zone (queue, aperture, post-exit) rather than aggregated into one count – across bottleneck, bidirectional, and crossing geometries. Two failure modes emerge: TTC anticipation does not break symmetric arching deadlocks at

Gang, P. and Veluri, A.
A Diagnostic Methodology for Failure-Mode Ablation of SFM, TTC, and ORCA Pedestrian Steering Paradigms.
DOI: 10.5220/0015190200004094
Paper published under CC license (CC BY-NC-ND 4.0)
In *Proceedings of the 16th International Conference on Simulation and Modeling Methodologies, Technologies and Applications (SIMULTECH 2026)*, pages 474-481
ISBN: 978-989-758-857-0; ISSN: 2184-2841
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narrow exits, and ORCA *reciprocal yielding* (each agent assuming its neighbour takes half the avoidance manoeuvre) depresses counterflow throughput. A density-modulated additive blend of the three paradigms is introduced as a mitigation probe, calibrated against Forschungszentrum Jülich (FZJ) pedestrian trajectories, with an explicit held-out out-of-distribution (OOD) calibration bound.

We make two contributions. The methodological contribution is the audit principle above, together with the $C_1{+}\varepsilon$ control protocol that instantiates it: a paradigm ablation is paired with a minimum-intervention control (per-step Gaussian velocity noise at standard deviation σ) and the noise threshold at which the control recovers the paradigm's headline advantage ($\sigma_{50} = 0.049$ m/s, 95% CI $[0.044, 0.053]$ in our case) is reported as a ceiling on the paradigm-specific attribution. The case-study contribution is the zone-decomposed SFM/TTC/ORCA failure-mode ablation itself. Code, data, and reproduction scripts will be released on acceptance.

2 RELATED WORK

The Social Force Model (SFM) of Helbing and Molnár (Helbing and Molnár, 1995), extended with body compression and sliding friction (Helbing et al., 2000), remains the workhorse of pedestrian dynamics; velocity-obstacle methods formalised by Fiorini and Shiller (Fiorini and Shiller, 1998) culminate in Optimal Reciprocal Collision Avoidance (ORCA) (van den Berg et al., 2011), which solves a 2D linear program (LP) for the velocity closest to the preferred velocity under mutual avoidance; and Karamouzas et al. (Karamouzas et al., 2014) derived an anticipatory inverse-power-law time-to-collision (TTC) force from human trajectory statistics. Guy et al. (Guy et al., 2010) provide the effort-minimisation conversion of LP-selected velocities to forces that we reuse. These paradigms are complementary in principle but typically deployed individually; recent surveys of microscopic crowd-simulation algorithms document this persistent paradigm fragmentation across the field. Hybrids that combine at most two paradigms via discrete scenario-dependent switching have been reported (van Toll et al., 2021; Dang et al., 2024); switch points must themselves be calibrated and transition behaviour is often numerically brittle. Fruin's (Fruin, 1971) Level-of-Service framework places the LoS-E boundary at $\rho \approx 4$ ped/m^2, motivating the sigmoid centre we adopt for ORCA fade-out (§3.1); Steffen and Seyfried (Steffen and Seyfried, 2010) and the Mullick mirror-point construction (Mullick et al., 2024) establish Voronoi tessellation – each agent's density taken from the inverse area of its Voronoi cell (the floor region nearest that agent) – as a higher-fidelity local density estimator than fixed-disk grid counting at narrow throats, motivating our per-agent density signal. Crucially for the present work, we also find no prior study that audits a steering-paradigm mechanism claim against a minimum-intervention control: claimed paradigm-specific advantages have been reported observationally across this literature without interventions that would distinguish the paradigm's distinctive machinery from trivial confounds. Beyond force- and velocity-based steering, data-driven and neural predictors have advanced rapidly: generative models learn interaction patterns directly from trajectory corpora, and recent surveys contrast these learned approaches with knowledge-based simulators (Korbmacher and Tordeux, 2022; Dang et al., 2024). Recent microscopic work also pushes agents into the high-density regime, e.g. an all-densities model driven by dynamic interpersonal distances (Cristiani et al., 2023). Such advances broaden the model space but do not supply the mechanism-isolating controls our audit principle requires. The audit principle we propose (§1) addresses this absence directly.

3 METHODOLOGY

We describe (i) the density-modulated additive acceleration formulation that defines steering configurations C1–C4, (ii) the Voronoi-based density estimator driving the blending weight, and (iii) the phenomenological speed–density calibration.

3.1 Hybrid Steering Model

Each agent i is characterised by position $\mathbf{x}_i$, velocity $\mathbf{v}_i$, goal $\mathbf{g}_i$, radius r_i, desired speed v_i^0, mass m_i, and relaxation time τ_i. The per-agent steering is an *additive blend* of accelerations in which only the ORCA term is density-weighted:

$$\mathbf{a}_i = \mathbf{a}_i^{\text{des}} + \mathbf{a}_i^{\text{SFM}} + \mathbf{a}_i^{\text{TTC}} + w_o(\rho_i)\,\mathbf{a}_i^{\text{ORCA}} + \mathbf{a}_i^{\text{wall}}, \quad (1)$$

where $w_o(\rho_i) \in [0,1]$ fades ORCA out as local density rises (Eq. 2), and the desired, SFM, TTC, and wall components are unconditional. Per-component accelerations are obtained as $\mathbf{F}/m_i$ from the force expressions below. This yields four steering configurations: C1 (SFM only), C2 (SFM+TTC), C3 (SFM+ORCA), and C4 (the full blend). Algorithm 1 summarises one simulation timestep; Figure 1 shows the architecture schematically.

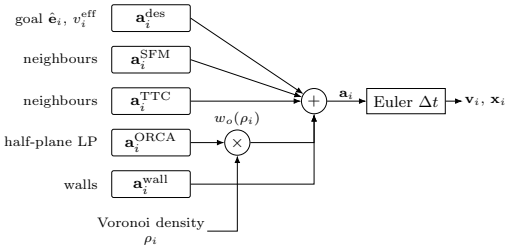


Figure 1: Density-modulated additive steering architecture (Eq. 1). Four acceleration components feed the accumulator unmodulated; the ORCA term is scaled by $w_o(\rho_i) = 1 - \sigma(\rho_i; \rho_0, k)$, with ρ_i the local Voronoi-cell density. The sum $\mathbf{a}_i$ drives the Euler integrator to update $(\mathbf{v}_i, \mathbf{x}_i)$.

Theoretical Positioning. Gating only ORCA reflects a separation between *regime-specific* and *always-on* contributions: SFM's contact term and TTC's anticipatory decay remain meaningful at every density, while ORCA presupposes a well-posed low-density velocity-selection problem that loses meaning at close packing. The LP-to-force conversion via a finite relaxation time is an engineering choice and does not preserve ORCA's collision-free guarantee.

The C_1 (SFM only) configuration also serves as the null against which the $C_1{+}\varepsilon$ minimum-intervention control is run (§4.6); comparing $C_1{+}\varepsilon$ to the ORCA-enabled C_3 and C_4 configurations, rather than comparing the ORCA configurations to each other, is what makes the audit informative.

Desired, SFM, and TTC Forces. The desired force follows Weidmann (Weidmann, 1993) speed–density coupling with $v_i^{\text{eff}}(\rho_i) = v_i^0(1 - e^{-\gamma(1/\rho_i - 1/\rho_{\max})})$; the pair $(\gamma, \rho_{\max})$ is calibrated against FZJ data (§3.3). Agent–agent interaction follows the Helbing et al. (Helbing et al., 2000) formulation with body-compression and sliding-friction terms; textbook parameters (A, B, k, κ) are in Table 1. Anticipatory repulsion uses the Karamouzas et al. (Karamouzas et al., 2014) inverse-power-law TTC force with gain k_{ttc} and decay timescale τ_0; τ_{ij} is set to $+\infty$ when no collision is predicted.

ORCA. ORCA (van den Berg et al., 2011) constructs half-plane velocity constraints from each neighbour and solves a 2D linear program for the velocity closest to the preferred velocity $\mathbf{v}_i^{\text{pref}} = v_i^0 \hat{\mathbf{e}}_i$. The LP-selected velocity is converted to a force $\mathbf{F}_i^{\text{ORCA}} = m_i(\mathbf{v}_i^{\text{ORCA}} - \mathbf{v}_i)/\tau_{\text{ORCA}}$ with $\tau_{\text{ORCA}} = 0.5$ s, following Guy et al. (Guy et al., 2010). This conversion is an engineering choice that allows ORCA to coexist with continuous-force paradigms inside Eq. (1), but it does not inherit ORCA's formal collision-free guarantee: that property is a feature of the LP-selected velocity itself and is not preserved by any force obtained through relaxation over a finite timescale.

Density-Dependent Weighting. The ORCA contribution is modulated by a sigmoid of local density:

$$w_o(\rho) = 1 - \sigma(\rho; 4.0, 2.0),\ \sigma(x; x_0, k) = \frac{1}{1+e^{-k(x-x_0)}}. \quad (2)$$

The fade centre $\rho_0 = 4.0\,\text{ped/m}^2$ is the Fruin LoS-E boundary (Fruin, 1971), above which pedestrians lose the freedom to choose their walking direction. We emphasise that this value is *not* fit to the empirical force-magnitude crossover $|\mathbf{F}_{\text{SFM}}| = |\mathbf{F}_{\text{ORCA}}|$, which occurs near $\rho \approx 0.07\,\text{ped/m}^2$ (§4.4) and measures a different quantity (raw force-magnitude equality versus paradigm applicability). The slope $k = 2.0$ yields a smooth transition of approximately $\pm 2\,\text{ped/m}^2$ around ρ_0. Only the ORCA term is density-modulated because SFM and TTC contribute body-compression/friction and anticipatory decay, respectively, that ORCA does not model.

Algorithm 1: One simulation timestep (density-modulated additive steering).

1: Build KDTree from active positions
2: $\mathcal{N}_i \leftarrow$ neighbours within $R = 3.0$ m
3: $\rho_i \leftarrow$ Voronoi cell density (§3.2)
4: $w_o(\rho_i) \leftarrow 1 - \sigma(\rho_i; 4.0, 2.0)$
5: $\mathbf{a}_i^{\text{des}} \leftarrow (v_i^{\text{eff}}(\rho_i)\hat{\mathbf{e}}_i - \mathbf{v}_i)/\tau_i$
6: $\mathbf{a}_i^{\text{SFM}} \leftarrow (1/m_i)\sum_{j\in\mathcal{N}_i} \mathbf{f}_{ij}$ ▷ SFM (Helbing et al., 2000)
7: $\mathbf{a}_i^{\text{TTC}} \leftarrow (1/m_i)\sum_{j\in\mathcal{N}_i} \mathbf{F}_{\text{TTC}}$ ▷ TTC (Karamouzas et al., 2014)
8: $\mathbf{v}_i^{\text{ORCA}} \leftarrow$ solve 2D LP from half-plane constraints
9: $\mathbf{a}_i^{\text{ORCA}} \leftarrow (\mathbf{v}_i^{\text{ORCA}} - \mathbf{v}_i)/\tau_{\text{ORCA}}$
10: $\mathbf{a}_i \leftarrow \mathbf{a}_i^{\text{des}} + \mathbf{a}_i^{\text{SFM}} + \mathbf{a}_i^{\text{TTC}} + w_o\,\mathbf{a}_i^{\text{ORCA}} + \mathbf{a}_i^{\text{wall}}$
11: $\mathbf{v}_i \leftarrow \mathbf{v}_i + \mathbf{a}_i\Delta t$; $\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i\Delta t$
12: Clamp $\|\mathbf{v}_i\| \le 2v_i^0$; deactivate agents at goal

3.2 Local Density via Voronoi Tessellation

The blending weight is driven by a per-agent local density estimate $\rho_V(i) = 1/A_i^{\text{Vor}}$, where A_i^{Vor} is the Voronoi cell area clipped to the scenario domain. Boundary cells are stabilised with the mirror-point construction of Mullick et al. (Mullick et al., 2024). Voronoi density is preferred over fixed-disk grid counting because grid averaging smooths over the local hotspots at narrow throats that drive paradigm switching (Steffen and Seyfried, 2010).

3.3 Phenomenological Calibration

The Weidmann coefficients $(\gamma, \rho_{\max})$ are the primary drivers of mismatch in the fundamental diagram (FD)

Table 1: Model parameters. $^{\dagger}\gamma$ and $\rho_{\max}$ refit via single-start Nelder–Mead ($x_{\text{atol}}=10^{-3}$, $f_{\text{atol}}=10^{-4}$); terminated at the seed-noise floor (53 evaluations, 3 seeds each). A, B fixed at textbook values (Helbing et al., 2000); a full 4-parameter multi-restart refit is deferred to future work (§6).

Parameter	Sym.	Default	Cal.†
Weidmann shape	γ	1.913	0.833
Jam density	$\rho_{\max}$	5.4	5.98
Social repulsion	A (N)	2000	– (fixed)
Repulsion range	B (m)	0.08	– (fixed)
Body compression	k (kg/s^2)	1.2×10^5	–
Friction	κ (kg/ms)	2.4×10^5	–
TTC gain / decay	k_{ttc}, τ_0	1.5, 3.0 s	–
ORCA horizon	τ_h (s)	5.0	–
Desired speed	v_0 (m/s)	1.34	–
Agent radius	r (m)	0.25	–
Timestep	Δt (s)	0.01	–

– the empirical speed–density relation a pedestrian model must reproduce. We fit $\phi = (\gamma, \rho_{\max})$ with A, B fixed at textbook Helbing et al. (Helbing et al., 2000) values via Nelder–Mead against binned FZJ measurements $\{(\rho_k, v_k^{\text{emp}})\}$:

$$\phi^* = \arg\min_{\phi} \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(v_k^{\text{sim}}(\phi) - v_k^{\text{emp}}\right)^2}. \qquad (3)$$

Optimisation terminates at the seed-noise floor (evaluations 44–53 oscillate within ± 0.003 m/s). Calibrated values $\gamma^* = 0.833$ and $\rho_{\max}^* = 5.98$ reduce in-sample FD RMSE from 0.235 to 0.099 m/s (§4.2).

Metrics. A *contact-overlap* is counted each timestep in which two agents satisfy $d_{ij} < r_i + r_j$; the metric is timestep-dependent by construction (halving Δt approximately doubles the per-run sum) and we use it for relative comparison at fixed $\Delta t = 0.01$ s. As a Δt-independent supplement we report *contact-events* (maximal runs of consecutive overlap timesteps per pair; §4.6). Robustness to $\Delta t \in \{0.005, 0.01, 0.02\}$ s is verified: the C1 > C4 rank ordering holds at all three values, while absolute counts scale inversely with Δt as expected. For scenarios that do not fully evacuate within the time window we report *throughput* rather than evacuation time; deadlock success is a binomial proportion with 95% Wilson CI. Table 2 summarises the four scenarios.

Table 2: Scenario summary.

Scenario	Geometry	N	Purpose
Corridor	18×5 m, periodic	9–162	FD calibration
Bottleneck	10×10 m, var. exit	50–100	Evac., deadlock
Bidirectional	20×3.6 m	200	Counter-flow
Crossing	90°, corridors	200	Intersection

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

We evaluate C1 (SFM only), C2 (SFM+TTC), C3 (SFM+ORCA), and C4 (full hybrid). Agents are heterogeneous with speed drawn from $\mathcal{N}(1.34, 0.26)$ m/s and radius from $\mathcal{N}(0.25, 0.03)$ m. ORCA's per-agent linear program makes C3/C4 approximately 40× slower than C1/C2 at 200 agents, imposing a practical budget of $n = 25$ replications per cell (ablation and C3/C4 of the FD; C1/C2 of the FD use $n = 30$ from an earlier run). Statistical tests are negative-binomial GLM (NB GLM) for contact-overlap counts (dispersion estimated by method of moments, with a duration offset for per-run rates), Cox proportional hazards for deadlock survival with Fisher's exact for the C2 complete-separation case, and linear mixed models (LMM) with random intercept by seed for continuous metrics. Holm–Šidák correction is applied across pairwise configuration comparisons within each scenario × metric combination to control the family-wise error rate at $\alpha = 0.05$; paired tests are used wherever seeds match across configurations. Reproduction scripts, parameter files, and raw results, with a sha256 artefact manifest, will be released on acceptance.

4.2 FD Calibration

Calibration against 4 776 FZJ unidirectional-corridor points reduces in-sample speed–density RMSE from 0.235 to 0.099 m/s (Figure 2); the FD is insensitive to steering configuration, so this alignment is phenomenological and does not validate the paradigm comparison.

4.3 Held-out OOD Validation

On held-out FZJ bottleneck geometries, the calibrated model systematically underestimates flow rates by 26–43% across all exit widths (Table 3); we report this gap as an explicit out-of-distribution calibration bound. A per-paradigm decomposition (not shown) confirms the bias is approximately uniform across C1–C4 (≤ 2 pp spread), so relative comparisons between configurations are invariant to the absolute offset and all quantitative ablation conclusions in §§4.5–4.6 survive it.

4.4 Density-Adaptive Gate

All observed densities in our benchmark geometries remain below the gate centre $\rho_0 = 4.0$ ped/m^2 (99.9th

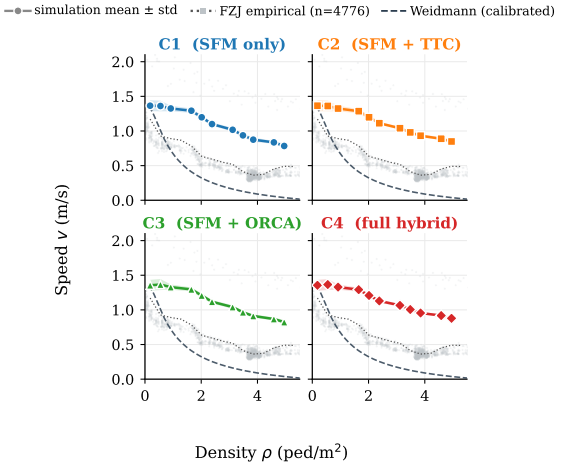


Figure 2: Speed–density fundamental diagram, separated by steering configuration (**C1**: SFM only; **C2**: SFM+TTC; **C3**: SFM+ORCA; **C4**: full hybrid). Coloured curves: simulation mean (line) with ± std band (shaded), bin centres marked. Grey: 4 776 FZJ empirical points with their binned mean (dotted). Dashed: calibrated Weidmann reference. All four configurations yield nearly indistinguishable trends; the FD is determined by the Weidmann term, not by steering choice. Calibration reduces in-sample RMSE from 0.235 to 0.099 m/s.

Table 3: Held-out OOD validation: simulated vs. FZJ empirical bottleneck flow rates. All relative errors are negative (systematic underestimation).

w (m)	J_{emp} (ped/s)	J_{sim} (ped/s)	σ_{sim}	Abs. err.	Rel. err.
2.4	5.125	3.547	0.098	-1.577	-30.8%
3.0	5.812	4.272	0.106	-1.540	-26.5%
3.6	6.939	4.592	0.169	-2.348	-33.8%
4.4	7.789	5.101	0.182	-2.688	-34.5%
5.0	9.219	5.266	0.221	-3.952	-42.9%

percentile $\approx 1.56\,\text{ped/m}^2$), so the sigmoid operates on its low-density branch throughout and the gate functions as a near-fixed blend rather than an adaptive transition in this study; the literature-motivated ρ_0 is retained over the force-magnitude crossover at $\rho \approx 0.07\,\text{ped/m}^2$ (§3.1), which measures a different quantity. Scenarios that traverse the transition band, together with a sensitivity sweep over (ρ_0, k), are flagged for future work (§6).

4.5 Paradigm Ablation across Scenarios

We compare C1–C4 across three scenarios at $n = 25$ per cell. Bottleneck uses 50 agents through variable exit widths; bidirectional uses 100+100 counter-flowing agents in a corridor; crossing uses 100+100 agents at a 90° intersection. Bottleneck contact-overlaps are classified by spatial zone relative to the exit at $x = 10\,\text{m}$: upstream ($x < 8$, queue formation), throat ($8 \le x \le 10.5$, the aperture itself), and downstream ($x > 10.5$, post-exit). This classification separates mechanisms that aggregate metrics would otherwise conflate.

TTC Eliminates Persistent Throat Overlaps at Narrow Bottlenecks. Persistent contact-overlaps during the exit-throat squeeze are rare events: at $w = 1.0\,\text{m}$, C1 produces 1.20 ± 1.96 overlap-timesteps per run whereas TTC-containing configurations (C2, C4) produce zero (paired Wilcoxon $p = 0.011$; Table 4, Figure 3). C3 (ORCA without TTC) yields 0.80 ± 2.87: ORCA's reciprocal yielding produces transient aperture-squeeze overlaps that TTC's anticipatory braking avoids. The pattern persists at $w \in \{1.2, 1.8\}\,\text{m}$ (C1 0.80, 0.32; C3 0.76, 0.56; C2 and C4 identically zero) and vanishes at $w \ge 2.4\,\text{m}$, so the mechanism is specific to narrow exits. All overlaps localise to the throat band ($8 \le x \le 10.5\,\text{m}$); upstream agents have room to manoeuvre and do not persistently overlap.

Methodological Note. The overlap counts use rejection-sampled placement (no pairwise overlap at $t = 0$); an earlier pipeline counting placement-phase overlaps produced a signal dominated (∼100%) by spawn artefacts resolving within ∼40 ms. Evacuation times are unaffected ($|\Delta t_{evac}| \le 0.5\,\text{s}$).

Table 4: Mean contact-overlaps by spatial zone ($n = 25$, mean ± std).

Config	Width (m)	Upstream	Throat	Downstream
C1	0.8	320.9 ± 55.3	1.8 ± 1.9	0.0 ± 0.0
C1	1.0	41.8 ± 15.8	0.6 ± 1.7	0.0 ± 0.0
C4	0.8	215.0 ± 54.3	0.6 ± 1.3	0.0 ± 0.0
C4	1.0	29.3 ± 12.2	0.1 ± 0.4	0.0 ± 0.0

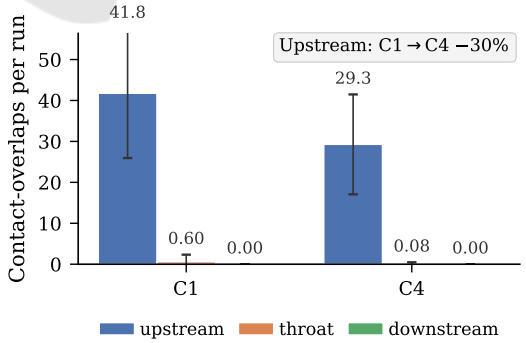


Figure 3: Contact-overlaps by zone at $w = 1.0\,\text{m}$ ($n = 25$, mean ± std) after spawn-artefact removal (§4.5). Persistent overlaps are rare and localise in the throat band (exit aperture); upstream and downstream zones are empty in all configurations. C2 and C4 (TTC-containing) are identically zero across every zone.

Crossing Scenario. At the 90° intersection, SFM-only (C1) agents reach force-balance standoffs (opposing repulsive forces cancel, leaving agents stationary) at the crossing centre, with only 5.9 of 200 agents exiting within the 60 s window. C2 roughly triples throughput (16.1/200; LMM coef. +10.16, ratio-of-means IRR = 2.72, $p_{\mathrm{HS}} < 10^{-40}$) and C4 matches or slightly exceeds C2 (17.1/200; IRR = 2.89, $p_{\mathrm{HS}} < 10^{-48}$). ORCA alone (C3) yields no gain (6.4/200; p_{HS}=0.95): the predictive component of TTC, not ORCA's LP-selected avoidance velocity, is what dissolves the force-balance standoff. The 90° intersection is pathological for force-balance models; we read this as relative mitigation within a scenario where C1 performs poorly, not a claim of strong absolute intersection performance.

Bidirectional Scenario. In counter-flowing groups ($n = 25$), C4 increases mean speed over C1 (LMM coef. +0.074 m/s, 95% CI $[0.048, 0.100]$, $p_{\mathrm{HS}} < 10^{-6}$) at comparable throughput (coef. +4.76; p_{HS}=0.60). ORCA alone (C3) reduces throughput (coef. −5.0); C4 recovers it because TTC dampens the gridlock pure ORCA yielding precipitates – a second instance of complementarity.

Oracle Baseline. Against a *per-scenario best-paradigm baseline* (the single C1/C2/C3 best on each metric), C4 improves on the oracle in 5 of 8 scenario–metric pairs, matches within 1% in 2, and underperforms by 1.4% in 1 ($w = 3.6$ m evacuation time, saturated regime). C4 avoids worst-case paradigm selection without requiring runtime scenario classification.

4.6 Bottleneck Deadlock and the C_1+ε Control

Table 5 reports mean evacuation times ($n = 25$) across four exit widths, with C1 vs. C4 differences tested by Holm–Šidák-corrected LMM. The hybrid produces a statistically significant evacuation-time reduction at narrow exits (9.3% at $w = 1.0$ m, 7.4% at $w = 1.2$ m, converging by $w \geq 2.4$ m; Table 5, Figure 5).

Narrow-Exit Deadlock ($w = 0.8$ m). At a 0.8 m exit with 100 agents ($n = 25$, seeds 42–66), completion rates are C1: 0/25, C2: 0/25, C3: 13/25 (52%), C4: 14/25 (56%). Neither plain SFM nor SFM+TTC resolved the $w = 0.8$ m arch within the 600 s window; both ORCA-enabled configurations resolved roughly half of trials.

The qualitative gap between $\{C1, C2\}$ and $\{C3, C4\}$ is therefore categorical in the sense of Fisher's exact test (complete separation), which is itself diagnostic: a simulator choice fully determines whether the crowd evacuates.

σ-Sweep Mechanism Study. To separate geometric symmetry-breaking from ORCA's velocity-space optimisation, we ran a σ-sweep of C_1+ε (plain SFM with per-step isotropic Gaussian velocity noise of standard deviation σ) over $\sigma \in \{0, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.10, 0.20\}$ m/s, $n = 25$ seeds per σ. Completions were $\{0, 1, 3, 9, 13, 18, 23, 24, 25\}$, yielding a well-defined monotone transition; a logistic fit gives $\sigma_{50} = 0.049$ m/s (95% CI $[0.044, 0.053]$), i.e. the σ at which fitted completion reaches 50%. The noise-vs-ORCA attribution ratio at $\sigma = 0.05$ is 1.00 (95% CI $[0.54, 1.83]$): within $n = 25$, passive Gaussian noise at $\sigma \approx \sigma_{50}$ is indistinguishable from ORCA's LP-based velocity selection as a deadlock-breaking mechanism (Figure 4); the residual velocity-space-optimisation contribution of ORCA is below our sampling resolution and larger-n separation is future work.

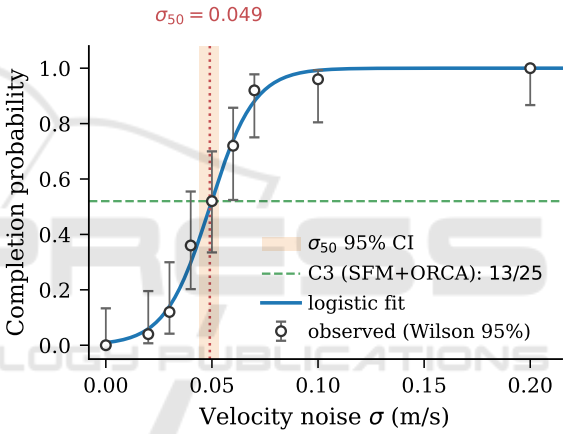


Figure 4: C_1+ε deadlock-resolution probability vs. Gaussian velocity noise σ ($w = 0.8$ m, 100 agents, $n = 25$); C_1+ε is plain SFM (**C1**) with per-step isotropic Gaussian velocity noise added to each agent. Blue: logistic fit. Vertical dotted line: $\sigma_{50} = 0.049$ m/s, 95% CI $[0.044, 0.053]$ (shaded). Horizontal dashed line: completion rate of the SFM+ORCA configuration **C3** (13/25) as a reference.

The Cox proportional hazards model (Holm–Šidák-corrected across the family {C1 vs. C2, C1 vs. C3, C1 vs. C4}) is consistent with the contingency-table result: C3 vs. C1 HR = 17.2, p_{HS}=0.048; C4 vs. C1 HR = 19.6, p_{HS}=0.044; C2 shows complete separation and is reported via Fisher's exact.

Arch Lifetime. Across the 25 non-evacuating C1 seeds at $w = 0.8$ m, the median terminal stall is 457 s with a mean of 3.2 agents frozen at the throat (24/25 seeds sustain > 100 s); trajectories show the stuck agents fully stationary (Figure 6), confirming that arches are force-balance configurations SFM cannot resolve passively. The 600 s window provides horizon-limited lower bounds on stall duration.

Table 5: Bottleneck evacuation time and contact-overlaps (mean ± std, $n = 25$; rejection-sampled initial placement, §4.5). Evacuation time: ** $p_{HS} < 0.01$; * $p_{HS} < 0.05$ for C1 vs. C4 via LMM. Contact-overlap counts after removing spawn-placement artefacts are rare events; C2 and C4 (TTC-containing) are identically zero at every tested width.

Evacuation time (s)				
w (m)	**C1**	**C2**	**C3**	**C4**
1.0	56.5±5.5	54.7±5.2	50.3±3.7	**50.8±4.4****
1.2	38.6±2.7	38.7±2.9	36.2±3.1	**36.1±2.5****
1.8	21.4±1.2	21.3±1.2	20.6±1.1	**20.7±1.0***
2.4	16.4±0.9	16.3±0.8	16.1±0.8	**16.1±0.9**
Contact-overlap count (persistent queue overlaps)				
1.0	1.2±2.0	**0.0±0.0**	0.8±2.9	**0.0±0.0**
1.2	0.8±1.8	**0.0±0.0**	0.8±1.9	**0.0±0.0**
1.8	0.3±1.0	**0.0±0.0**	0.6±2.8	**0.0±0.0**
2.4	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0

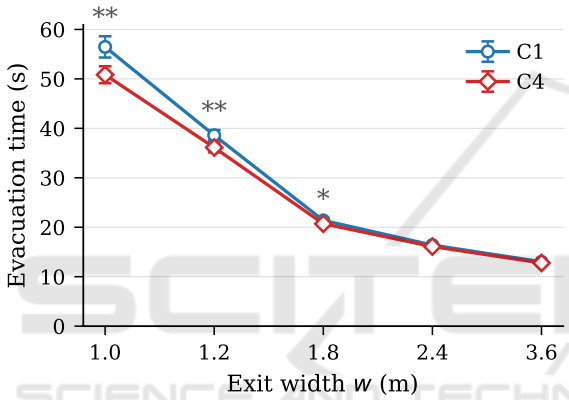


Figure 5: Evacuation time vs. exit width for C1 and C4. Error bars: 95% CIs, $n = 25$.

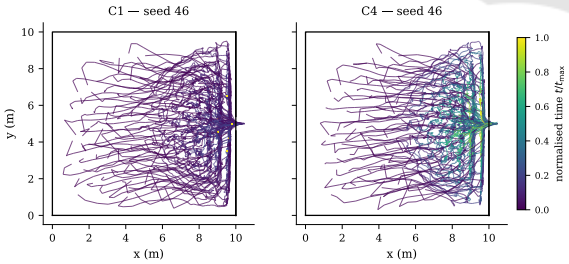


Figure 6: Agent trajectories at $w = 0.8$ m (100 agents, seed 46). **Left:** C1 forms a stable arching deadlock at the exit throat. **Right:** C4 destabilises the arch and evacuates all 100 agents. Colour: normalised time.

5 DISCUSSION

Why Each Component Matters, and Why Each Can Also Harm. TTC and ORCA play distinct, complementary roles, each harmful in the wrong regime. TTC drives exit-throat overlaps to zero and triples crossing throughput, yet cannot break symmetric arches (C2: 0/25 at $w = 0.8$ m); ORCA-enabled configurations resolve the arch (13–14/25) but depress counterflow, which C4 recovers because TTC dampens the gridlock pure ORCA yielding precipitates. Naïve always-on superposition can thus be *worse* than any single paradigm, motivating the density-gated architecture.

The Audit Principle as Theoretical Contribution. The C_1+ε control changes what we can conclude from the SFM/TTC/ORCA ablation: within our sampling resolution, ORCA's arch-resolution advantage over SFM is explained by passive symmetry-breaking rather than LP-based velocity-space optimisation, so the claim "ORCA resolves arching because of its distinctive machinery" does not survive the audit at $n = 25$ even though "ORCA resolves arching" does. Generalised, the C_1+ε protocol (§1) is a transferable template: pair a paradigm ablation with an additive-noise variant of the baseline and report the noise level matching the paradigm's headline advantage (σ_{50}) as a ceiling on the paradigm-specific attribution.

Implications for Practitioners. Plain SFM underestimates narrow-exit evacuation safety categorically (0/25 vs. 14/25 at $w = 0.8$ m), so SFM-based safety analyses should be audited for symmetry-breaking artefacts; single-paradigm simulators mispredict neighbouring scenario classes (TTC alone mispredicts deadlock, ORCA alone counterflow); and the 26–43% FD-only underestimation is a bias no steering choice closes, so users should expect a geometry-specific calibration step before trusting absolute predictions. A qualitative JuPedSim v1.3.2 comparison at a 1.0 m bottleneck independently confirms velocity-based models evacuate faster than force-based ones at moderate widths.

Limitations. $n = 25$ limits attribution power for the C_1+ε control; a larger-n sweep is the most important next step. Calibration is in-sample on FZJ corridor data; a 4-parameter multi-restart refit with bottleneck flow in the loss remains future work. The density-adaptive gate is not exercised in its transition regime here, functioning as a near-fixed blend. ORCA's per-agent LP adds ∼30–40× overhead, making C3/C4 real-time-ineligible beyond ∼30 agents without C/C++.

6 CONCLUSIONS

We have argued that failure-mode ablations of pedestrian steering paradigms should be accompanied by minimum-intervention controls, and instantiated this with the C_1+ε protocol on the SFM/TTC/ORCA

triad–the audit principle and protocol being the methodological contribution, the ablation its first application. The control locates $\sigma_{50} = 0.049$ m/s, at which passive Gaussian noise is indistinguishable from ORCA's LP-based selection as a deadlock-breaker ($n = 25$), so ORCA's arch-resolution advantage is attributable to symmetry-breaking rather than distinctive machinery. The ablation further shows TTC and ORCA each harmful off-regime (TTC cannot break arches at $w = 0.8$ m; ORCA depresses counterflow), while a held-out 26–43% flow underestimation is uniform across C1–C4, so FD alignment does not validate steering. These findings favour failure-mode ablation with minimum-intervention controls and honest OOD reporting over aggregate benchmarking; gating only the regime-dependent paradigm is a blend architecture that follows from the same diagnosis.

Future work. Beyond the larger-n σ-sweep, two directions follow: applying the $C_1{+}\varepsilon$ audit protocol to hybrid-paradigm claims beyond this triad (e.g., SPH-coupled and cognitive-heuristic hybrids), and trajectory-level validation on ETH–UCY and further FZJ scenarios that exercise the adaptive transition region.

ACKNOWLEDGMENTS

The author used Anthropic's Claude Code as a development aid for code generation, debugging, and reference fact-checking. Claude was *not* used for research ideation, experimental design, manuscript writing, or the generation of results, analyses, or figures. All AI-assisted code was reviewed and verified by the author, who takes full responsibility for this manuscript in accordance with COPE. The authors further extend their gratitude to Ninaad Adhvaryu for his countless reviews and feedback. We move on to pay respects to the 3.2 agents who, on average, remain frozen at the exit throat to this day. Last of all, we thank our friends and family, whose patience comfortably outlasted the median 457 s arch, and who—unlike our agents—could have left the room at any time but chose to stay.

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